

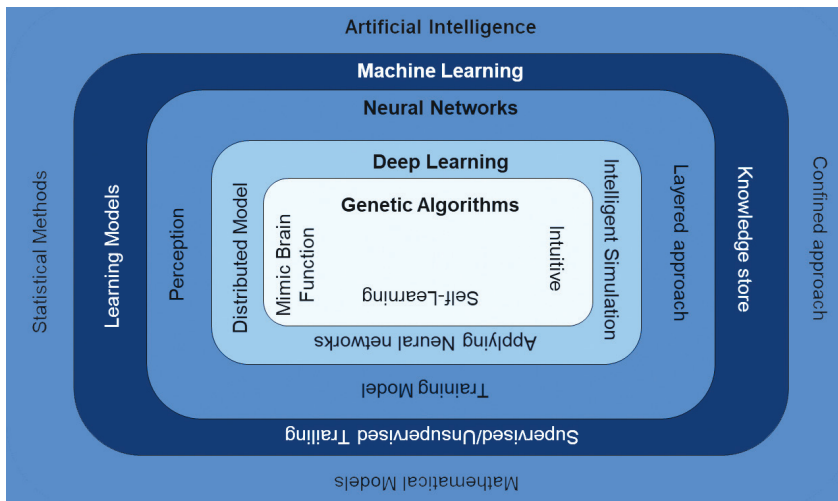


Figure 1: The paradigm of AIML

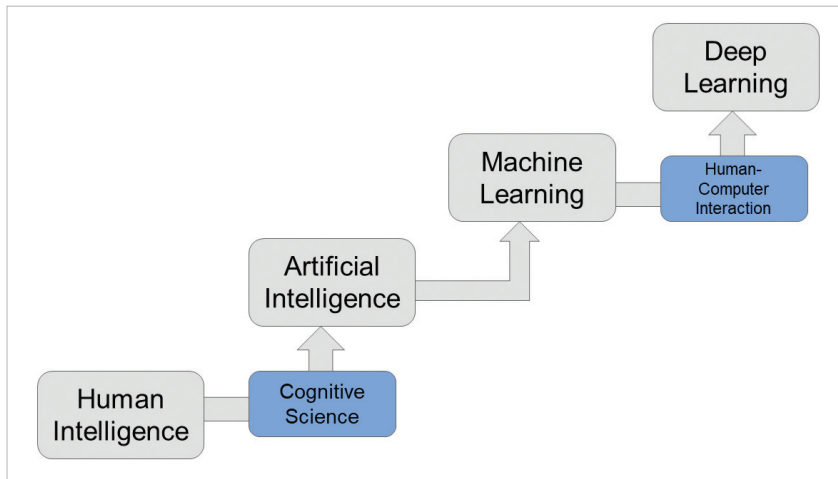


Figure 2: Singularity of human intelligence to deep learning

Importance of tracking AI trends

The importance of tracking AI trends can be summarised in the following points.

Stay ahead of the competition:

AI is rapidly evolving, and businesses aiming to remain competitive need to be aware of the latest trends. By tracking AI trends, businesses can identify new opportunities and threats, and develop strategies to stay ahead.

Identify new market opportunities:

AI is being used in a wide variety of industries, with new applications emerging all the time. Tracking AI trends helps businesses identify untapped market opportunities.

Make better strategic decisions: AI can be used to collect and analyse data, identify patterns, and make predictions. Businesses can use this information to make better strategic decisions about their products, services, and operations.

Improve efficiency and productivity: AI can automate tasks, improve decision-making, and optimise processes. By keeping an eye on AI trends, businesses can find ways to improve their efficiency and productivity.

Reduce costs: AI can automate tasks currently performed by humans, resulting in cost savings. Tracking AI trends enables businesses to identify cost-saving opportunities.

Improve customer experience: AI can personalise customer experiences, provide better customer service, and predict customer needs. Tracking AI trends helps businesses improve their customer experience.

Explainable AI

Explainable artificial intelligence (XAI) refers to the development of AI systems that can provide understandable and transparent explanations for their decision-making processes. XAI is essential because traditional AI models, like deep neural networks, often operate as ‘black boxes’, making it challenging to comprehend why they reach specific conclusions or predictions.

XAI aims to bridge this gap by enabling humans to understand and trust AI systems. It involves techniques such as model interpretability, feature importance analysis, and rule-based systems to make AI more transparent. This transparency enhances accountability, aids in debugging and improving models, and ensures that AI systems adhere to ethical and regulatory standards.

Various approaches to XAI exist, with the best approach depending on the specific AI model and application.

Feature importance: Identifies the most important features for an AI model’s decision-making

Counterfactual explanations: Shows how an AI model’s prediction would change if input features were different

Local explanations: Explains an AI model’s decision for a specific input example

Model introspection: Allows users to explore an AI model’s internal workings

XAI is a rapidly growing field with ongoing research, and has the potential to make AI systems more transparent, trustworthy, and fair.

As an example, a hospital can use XAI to explain how a machine learning model is making decisions about cancer treatment. This can help doctors to understand the model’s reasoning and